No One Fits All: From Fixed Prompting to Learned Routing in Multilingual LLMs

Wei-Chi Wu[1], Sheng-Lun Wei[1], Hen-Hsen Huang[2], Hsin-Hsi Chen[1]
[1]National Taiwan University [2]Academia Sinica, Taiwan

TL;DR: **No single prompting strategy fits all languages. We train a lightweight classifier that picks native or translation prompting for each instance. It beats every fixed strategy, and shows that resource level, not translation quality, decides when translation helps.**

Fixed Prompting → Learned Routing

- Translation-based prompting often boosts multilingual LLMs, but native prompting can sometimes win.
- **Experimental Setup:**
  - **Models:** DeepSeek-v3.1, Llama-3.3-70B
  - **10 languages** by resource level: High (ZH, ES, DE, HI) · Mid (BN, ID, KO) · Low (SI, SW, YO).
  - **Datasets**:
    In-domain: Global-MMLU and MMLU-ProX
    Out-of-domain: XQuAD, mCSQA, and XCOPA

RQ1: No Single Strategy Fits All

- **Six strategies** (all zero-shot CoT)
  - **NATIVE**: using the native language, no translation
  - **TRANSLATE**: translate the question to English first
  - **SEL-TRANS**: native question, but English prompting
  - **SCoT-NATIVE** / **SCoT-TRANS**: add a strategic reasoning structure, native or translate to English
  - **PROMPT-ROUTING**: model decides native or translate
- **Findings**
  - No single strategy dominates
  - Resource level predict strategy effectiveness
  - Prompt-based strategy selection fails

Prompt Method	ZH	ES	DE	HI	BN	ID	KO	SI	SW	YO	Avg
Native	88.2	89.4	88.2	84.6	83.1	88.0	36.7	75.3	75.5	45.2	75.4
Translate	87.5	89.0	88.4	**86.2**	85.8	**88.4**	**86.6**	**82.7**	**81.0**	64.0	84.0
Sel-Trans	**88.8**	89.3	88.6	85.7	85.4	88.2	**86.6**	80.8	79.6	**64.4**	83.7
SCoT-Native	85.8	86.5	84.3	83.2	71.3	82.5	77.6	71.0	65.2	46.4	75.4
SCoT-Trans	88.3	**89.5**	**88.9**	85.7	**87.1**	88.3	86.4	81.1	80.9	63.5	**84.0**
Prompt-Routing	87.5	89.2	88.2	85.5	85.2	88.0	78.5	81.7	80.9	62.8	82.8

RQ3: Why Low-Resource Languages Favor Translation

- Low-resource responses concentrate in low-quality translation bins, yet they benefit the most from translation.
- As translation quality rises, all methods improve and the gap between strategies narrows.
- The trained classifier favors translation most strongly where translation quality is lowest.
- **Language resource level, not translation quality, decides the optimal strategy.**

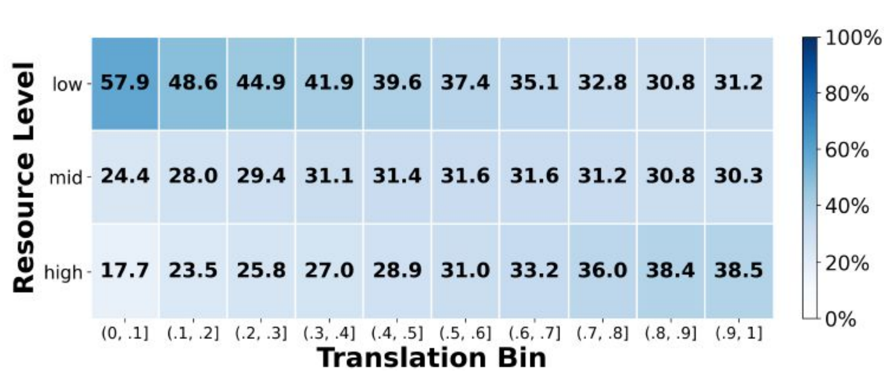
	Native		*Translate*		*Classifier*		*Gap (T-N)*		*Trans Rate (%)*	
Quality Percentile	**DS**	**Llama**	**DS**	**Llama**	**DS**	**Llama**	**DS**	**Llama**	**DS**	**Llama**
10%	53.2	50.3	67.3	57.6	69.3	58.2	14.2	7.3	83.5	76.3
20%	57.4	52.2	71.4	59.7	72.8	60.4	14.0	7.5	81.9	72.0
30%	60.3	55.2	73.7	62.3	74.8	63.2	13.5	7.1	80.1	67.1
40%	62.1	57.7	75.0	64.4	75.9	65.3	12.9	6.7	78.6	63.2
50%	64.1	59.7	76.3	66.1	77.1	66.9	12.2	6.4	77.6	60.3
60%	65.8	61.3	77.4	67.4	78.1	68.3	11.6	6.0	76.6	58.0
70%	67.5	62.9	78.5	68.7	79.2	69.6	11.0	5.8	76.0	56.0
80%	69.2	64.3	79.5	69.8	80.1	70.7	10.3	5.6	75.5	53.8
90%	70.8	65.7	80.5	70.9	81.1	71.9	9.8	5.2	75.2	51.6
100%	72.5	67.2	81.7	72.1	82.3	73.1	9.3	4.9	75.2	49.2

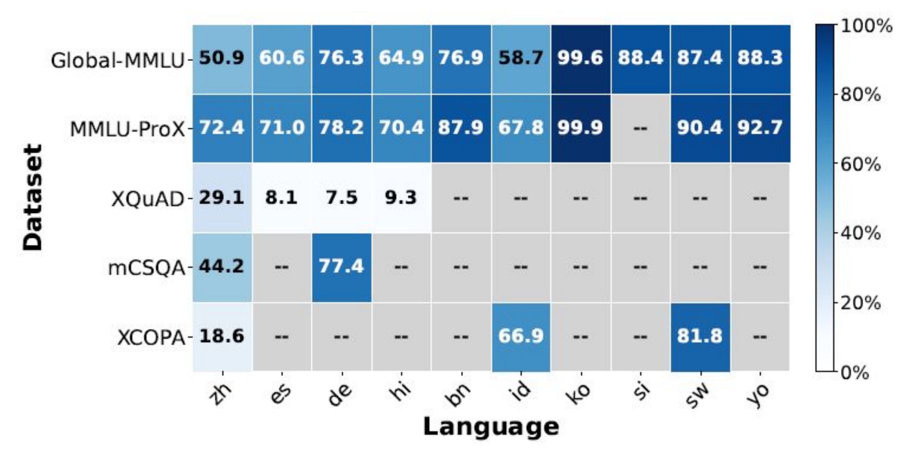
RQ2: Can We Learn to Select?

- We frame strategy choice as a binary decision between **NATIVE** and **TRANSLATE**.
- We **train lightweight classifiers**, XGBoost and MLP, to learn the decision.
- The classifier beats every fixed strategy, reaching 82.3 on Global-MMLU with +0.6 over Translate and +9.8 over Native.
- It transfers to unseen task types, scoring 87.6 on XQuAD, 95.7 on XCOPA, and 33.8 on mCSQA.
- The **gains are statistically significant**, with p < 0.001 for XGBoost.

		High-Resource				*Mid-Resource*			*Low-Resource*			
Dataset	**Method**	**ZH**	**ES**	**DE**	**HI**	**BN**	**ID**	**KO**	**SI**	**SW**	**YO**	**Avg**
Global-MMLU	Native	86.4	86.4	84.6	83.1	79.9	85.3	36.1	72.0	71.7	39.0	72.5
	Translate	86.0	86.8	85.5	84.4	83.0	86.0	84.8	80.0	79.2	61.6	81.7
	Classifier (Ours)	**86.8**	**87.3**	**86.0**	**84.7**	**83.3**	**86.4**	**84.9**	**80.0**	**79.8**	**63.7**	**82.3**
	Oracle	90.5	90.6	91.2	89.1	88.4	90.8	89.5	86.5	85.3	72.8	88.3
MMLU-ProX	Native	80.5	80.8	79.7	77.9	75.4	80.1	35.7	–	69.3	43.4	69.2
	Translate	80.3	81.0	80.3	79.0	78.5	80.5	**79.5**	–	75.6	64.5	77.7
	Classifier (Ours)	**80.8**	**81.3**	**80.5**	**79.2**	**78.7**	**80.7**	**79.5**	–	**76.0**	**64.6**	**77.9**
	Oracle	85.4	85.5	85.0	84.2	83.3	85.5	82.8	–	80.6	70.8	82.6
XQuAD	Native	86.6	87.2	89.2	83.6	–	–	–	–	–	–	86.7
	Translate	87.2	**88.2**	**90.6**	82.5	–	–	–	–	–	–	87.1
	Classifier (Ours)	**88.6**	87.9	89.2	**84.7**	–	–	–	–	–	–	**87.6**
	Oracle	91.0	92.0	94.1	89.8	–	–	–	–	–	–	91.7
mCSQA	Native	27.6	–	38.2	–	–	–	–	–	–	–	32.9
	Translate	28.2	–	38.5	–	–	–	–	–	–	–	33.4
	Classifier (Ours)	**28.4**	–	**39.1**	–	–	–	–	–	–	–	**33.8**
	Oracle	36.8	–	45.7	–	–	–	–	–	–	–	39.9
XCOPA	Native	97.0	–	–	–	–	95.8	–	–	87.4	–	93.4
	Translate	**97.4**	–	–	–	–	**96.8**	–	–	91.8	–	95.3
	Classifier (Ours)	**97.4**	–	–	–	–	96.6	–	–	**93.0**	–	**95.7**
	Oracle	99.0	–	–	–	–	98.6	–	–	97.0	–	98.2

- Word-overlap features dominate importance, capturing the alignment between native and translated responses.
- **PROMPT-ROUTING** cannot access these signals by self-assessment, which is why it loses to the learned classifier.
- High-resource languages get balanced selection, while low-resource languages strongly favor translation, contradicting prior work.





	XGBoost	**MLP**
DeepSeek-v3.1	1. Word Overlap (34.38%) 2. Metadata (24.52%) 3. POS (9.97%)	1. Word Overlap (35.10%) 2. Response Quality (12.62%) 3. Question-Level (10.55%)
Llama-3.3-70B	1. Word Overlap (56.91%) 2. Question-Level (24.62%) 3. Response Quality (17.11%)	1. Word Overlap (36.78%) 2. POS (10.95%) 3. Question-Level (10.41%)

Conclusion & Discussion

- **Multilingual prompting becomes a learned decision problem rather than a fixed strategy.**
- **A lightweight per-instance router beats any single strategy and transfers to unseen tasks.**
- **Future work can explore stronger routers, hybrid strategies, and routing inside inference to remove the dual-generation cost.**

National Taiwan University 國立臺灣大學
NLP Lab
@alexwcwu @kevinweisl

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